

How an Inference Engine Works — smol_vllm

 Engine workflow diagram

Arc: observe → build → break → debug → prove

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2	Build it — stubs → green checkpoints
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All engines use `seed=0` for reproducible demos.

0 — Setup

```
In [26]: import inspect
import subprocess
import sys
from pathlib import Path

SEED = 0

_shadow = Path.cwd() / "smol_vllm"
if _shadow.is_dir():
    sys.path = [p for p in sys.path if Path(p).resolve() != _shadow.resolve()]
if str(Path.cwd()) not in sys.path:
    sys.path.insert(0, str(Path.cwd()))

# After sync: reinstall editable package into *this* kernel's Python
subprocess.run(
    [sys.executable, "-m", "pip", "install", "-q", "-e", "./smol-vllm"],
    check=True,
)
```

```

# Drop cached imports from a pre-sync kernel session (smol_vllm + lib adapted)
for name in list(sys.modules):
    if (
        name == "smol_vllm"
        or name.startswith("smol_vllm.")
        or name.startswith("lib.")
        or name == "agent_demo"
    ):
        del sys.modules[name]

from smol_vllm import LLMEngine, SequenceStatus
from smol_vllm.demo import (
    run_block_manager_checkpoint,
    run_build_allocate_checkpoint,
    run_build_append_checkpoint,
    run_exp1_continuous_batching,
    run_exp2_memory_pressure,
    run_exp3_prefix_sharing,
    run_exp4_throughput_scaling,
    run_exp5_fake_vs_real,
    run_scheduler_checkpoint,
    reproduce_preemption_livelock,
    measure_kv_bytes_per_token,
)
from smol_vllm import exercises
from lib.engine_state import show_engine_state

_params = inspect.signature(LLMEngine.__init__).parameters
assert "seed" in _params, (
    "Old smol_vllm still loaded – use Kernel → Restart, then re-run this cell"
)
print("smol_vllm imported OK (seed, preempt_guard, output_mode, timing available)"

```

smol_vllm imported OK (seed, preempt_guard, output_mode, timing available)

In [27]: `from lib.gpu_check import load_dotenv, print_gpu_report`

```

load_dotenv()
GPU_REPORT = print_gpu_report()
GPU_OK = GPU_REPORT["real_model_ready"]

```

=====
Class 5 GPU / Lambda readiness
=====

smol_vllm import	OK
torch	OK
CUDA available	OK
CUDA device	NVIDIA A100-SXM4-40GB
nvidia-smi GPU	NVIDIA A100-SXM4-40GB
transformers	OK
HF_TOKEN in env	OK
LAMBDA in .env	OK (Mac rsync/tunnel)
Real model ready	OK

→ Part D (GPU) cells can run. First load takes ~30s (TinyLlama).
=====

1 — Walk the diagram (one request by hand)

Follow the workflow diagram: request → waiting → allocate → prefill → decode → free.

```
In [28]: engine = LLMEngine(
    num_gpu_blocks=24,
    block_size=16,
    max_batch_size=4,
    enable_metrics=True,
    seed=SEED,
)
print(f"Engine: {engine.block_manager.num_blocks} blocks × {engine.block_manager.block_size} tokens/block")
show_engine_state(engine)
```

```
Engine: 24 blocks × 16 tokens/block
waiting=[] running=[] swapped=[] | free=24/24 blocks util=0.0%
```

```
In [29]: # Warm-up: allocate + free so later tables get scattered physical IDs (not in order)
for sid, ntok in [(90, 32), (91, 48)]:
    engine.add_request(list(range(ntok)), max_tokens=1)
while engine.scheduler.waiting or engine.scheduler.running:
    engine.step()
# Re-create clean engine for the walkthrough
engine = LLMEngine(num_gpu_blocks=24, block_size=16, max_batch_size=4, enable_metrics=True, seed=SEED)

engine.add_request(list(range(10, 25)), max_tokens=12)
engine.add_request(list(range(100, 130)), max_tokens=8)
engine.add_request(list(range(200, 212)), max_tokens=20)
show_engine_state(engine)
```

```
[metrics] step=1 prefill=8ms decode=0ms gen_tokens=2 tok/s=247 running=0 waiting=0 swapped=0 kv=[0.0%] 0% cpu=0%
waiting=[0, 1, 2] running=[] swapped=[] | free=24/24 blocks util=0.0%
```

```
In [30]: # Diagram: can_allocate? → allocate → block_table
for gid in [0, 1, 2]:
    ntok = engine.groups[gid].sequences[0].num_tokens
    ok = engine.block_manager.can_allocate(ntok)
    if ok:
        engine.block_manager.allocate(gid, ntok)
        print(f"group {gid}: can_allocate={ok} block_table={engine.block_manager.block_table}")
show_engine_state(engine)
```

```
group 0: can_allocate=True block_table=[0]
group 1: can_allocate=True block_table=[1, 2]
group 2: can_allocate=True block_table=[3]
waiting=[0, 1, 2] running=[] swapped=[] | free=20/24 blocks util=16.7%
```

```
In [31]: # Prefill + decode steps (full logs – watch queues move)
for step in range(1, 8):
    outputs = engine.step()
    show_engine_state(engine, step=step)
```

```

if outputs and all(o.finished for o in outputs):
    break

```

```

[metrics] step=1 prefill=6ms decode=0ms gen_tokens=3 tok/s=518 running=3 w
aiting=0 swapped=0 kv=[██████████] 33% cpu=4%
step 1 | waiting=[] running=[0, 1, 2] swapped=[] | free=16/24 blocks ut
il=33.3%
[metrics] step=2 prefill=0ms decode=15ms gen_tokens=3 tok/s=198 running=3
waiting=0 swapped=0 kv=[██████████] 38% cpu=0%
step 2 | waiting=[] running=[0, 1, 2] swapped=[] | free=15/24 blocks ut
il=37.5%
[metrics] step=3 prefill=0ms decode=15ms gen_tokens=3 tok/s=199 running=3
waiting=0 swapped=0 kv=[██████████] 42% cpu=0%
step 3 | waiting=[] running=[0, 1, 2] swapped=[] | free=14/24 blocks ut
il=41.7%
[metrics] step=4 prefill=0ms decode=15ms gen_tokens=3 tok/s=199 running=3
waiting=0 swapped=0 kv=[██████████] 42% cpu=2%
step 4 | waiting=[] running=[0, 1, 2] swapped=[] | free=14/24 blocks ut
il=41.7%
[metrics] step=5 prefill=0ms decode=15ms gen_tokens=3 tok/s=199 running=3
waiting=0 swapped=0 kv=[██████████] 46% cpu=0%
step 5 | waiting=[] running=[0, 1, 2] swapped=[] | free=13/24 blocks ut
il=45.8%
[metrics] step=6 prefill=0ms decode=15ms gen_tokens=3 tok/s=199 running=3
waiting=0 swapped=0 kv=[██████████] 46% cpu=0%
step 6 | waiting=[] running=[0, 1, 2] swapped=[] | free=13/24 blocks ut
il=45.8%
[metrics] step=7 prefill=0ms decode=15ms gen_tokens=3 tok/s=199 running=3
waiting=0 swapped=0 kv=[██████████] 46% cpu=0%
step 7 | waiting=[] running=[0, 1, 2] swapped=[] | free=13/24 blocks ut
il=45.8%

```

```

In [32]: # generate() convenience API
clean = LLMEngine(num_gpu_blocks=24, block_size=16, max_batch_size=4, enable
for tok in clean.generate(list(range(5, 25)), max_tokens=8):
    pass
show_engine_state(clean)
clean.metrics.print_summary()

```

```

[metrics] step=1 prefill=2ms decode=0ms gen_tokens=1 tok/s=481 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=2 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=3 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=4 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=5 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=6%
[metrics] step=6 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=7 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 8% cpu=0%
[metrics] step=8 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=0 w
aiting=0 swapped=0 kv=[██████████] 0% cpu=0%
waiting=[] running=[] swapped=[] | free=24/24 blocks util=0.0%

```

```

[metrics] summary:
  prompt_tokens_total=20 generation_tokens_total=8
  time_to_first_token_avg_ms=2 tpot_avg_ms=5 e2e_request_latency_avg_ms=40
  prompt_len_avg=20 (prefix tokens at start)
  prefill_latency_avg_ms=2
  decode_latency_avg_ms=5

```

2 — Build it

You've watched the engine allocate blocks. Now write the allocator. **The checkpoint is the spec.**

Open `smol_vllm/exercises.py` — run each cell below. **First run = RED (expected)**. Implement, re-run until green.

```

In [33]: # 2.1 allocate() – run first. FAIL until you implement exercises.allocate.
try:
    run_build_allocate_checkpoint(exercises.allocate)
except NotImplementedError:
    print("RED: implement allocate() in smol_vllm/exercises.py, then re-run

# Escape hatch (instructor pre-class / stuck):
# from smol_vllm.reference.exercises_ref import allocate as ref_allocate
# run_build_allocate_checkpoint(ref_allocate)

```

RED: implement allocate() in smol_vllm/exercises.py, then re-run this cell

```

In [34]: # 2.2 append_slot() – same pattern
try:
    run_build_append_checkpoint(exercises.append_slot)
except NotImplementedError:
    print("RED: implement append_slot() in smol_vllm/exercises.py, then re-r

```

```
# Escape hatch:
# from smol_vllm.reference.exercises_ref import append_slot as ref_append_slot
# run_build_append_checkpoint(ref_append_slot)
```

RED: implement `append_slot()` in `smol_vllm/exercises.py`, then re-run this cell

In [35]: *# 2.3 Built-in checkpoints (always run after build section)*

```
run_block_manager_checkpoint()
run_scheduler_checkpoint(seed=SEED)
```

```
=====
Checkpoint: BlockSpaceManager
=====
```

```
Allocated 3 seqs x 3 blocks: util=45%
After appends (16 each): util=60%
After free(seq 1): util=40%
Free blocks: 12 (4 blocks returned from seq 1)
PASS: BlockSpaceManager checkpoint
```

```
=====
Checkpoint: Scheduler (continuous batching)
=====
```

```
Submitting 10 requests, max_batch_size=4
[metrics] step=1 prefill=4ms decode=0ms gen_tokens=4 tok/s=974 running=4 waiting=6 swapped=0 kv=[██████████] 6% cpu=4%
step 1: running=4, waiting=6
[metrics] step=2 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=6 swapped=0 kv=[██████████] 6% cpu=0%
step 2: running=4, waiting=6
[metrics] step=3 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=6 swapped=0 kv=[██████████] 6% cpu=0%
step 3: running=4, waiting=6
[metrics] step=4 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=6 swapped=0 kv=[██████████] 6% cpu=0%
step 4: running=4, waiting=6
[metrics] step=5 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=0 waiting=6 swapped=0 kv=[██████████] 0% cpu=0%
step 5: running=0, waiting=6
[metrics] step=6 prefill=4ms decode=0ms gen_tokens=4 tok/s=968 running=4 waiting=2 swapped=0 kv=[██████████] 6% cpu=0%
step 6: running=4, waiting=2
[metrics] step=7 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=2 swapped=0 kv=[██████████] 6% cpu=3%
step 7: running=4, waiting=2
[metrics] step=8 prefill=0ms decode=28ms gen_tokens=4 tok/s=141 running=4 waiting=2 swapped=0 kv=[██████████] 6% cpu=1%
step 8: running=4, waiting=2
PASS: Scheduler checkpoint
```

3 — Experiments (predict → run)

Scan the logs for `[preempt]`, `swapped=`, and `running=` — there will be many lines; that's intentional.

Predict — Exp 1 (continuous batching)

20 requests, `max_batch_size=4`. **Predict:** peak `running`? Does `waiting` drain monotonically?

```
In [36]: expl_engine = run_expl_continuous_batching(seed=SEED)
```

=====

Experiment 1: Continuous Batching

=====

Submit 20 requests of different lengths

Columns: step | running | waiting | swapped | blocks_used (%)

```
[metrics] step=1 prefill=18ms decode=0ms gen_tokens=4 tok/s=221 running=4
waiting=16 swapped=0 kv=[██████████] 19% cpu=3%
  step  1 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 18.8%
[metrics] step=2 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 22% cpu=0%
  step  2 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 21.9%
[metrics] step=3 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 22% cpu=0%
  step  3 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 21.9%
[metrics] step=4 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 22% cpu=0%
  step  4 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 21.9%
[metrics] step=5 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 22% cpu=0%
  step  5 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 21.9%
[metrics] step=6 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step  6 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=7 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=2%
  step  7 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=8 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step  8 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=9 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step  9 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=10 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 10 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=11 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=3%
  step 11 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=12 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 12 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=13 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 13 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=14 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 14 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=15 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 15 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=16 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
  step 16 | running= 4 | waiting=16 | swapped= 0 | blocks_used= 25.0%
[metrics] step=17 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=16 swapped=0 kv=[██████████] 25% cpu=0%
```


[illegible]

[illegible]

[illegible]

[illegible]

```

waiting=0 swapped=0 kv=[██████████] 27% cpu=0%
  step 92 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 26.6%
  [metrics] step=93 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 27% cpu=0%
  step 93 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 26.6%
  [metrics] step=94 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 27% cpu=0%
  step 94 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 26.6%
  [metrics] step=95 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 27% cpu=0%
  step 95 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 26.6%
  [metrics] step=96 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 28% cpu=0%
  step 96 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 28.1%
  [metrics] step=97 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 28% cpu=0%
  step 97 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 28.1%
  [metrics] step=98 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 28% cpu=2%
  step 98 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 28.1%
  [metrics] step=99 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 30% cpu=0%
  step 99 | running= 4 | waiting= 0 | swapped= 0 | blocks_used= 29.7%
  [metrics] step=100 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=
0 waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
  step 100 | running= 0 | waiting= 0 | swapped= 0 | blocks_used= 0.0%

```

Done in 100 steps. All 20 requests finished.

```

[metrics] summary:
  prompt_tokens_total=992 generation_tokens_total=400
  time_to_first_token_avg_ms=841 tpot_avg_ms=20 e2e_request_latency_avg_ms
=1230
  prompt_len_avg=50 (prefix tokens at start)
  prefill_latency_avg_ms=20
  decode_latency_avg_ms=20

```

Predict — Exp 2 (memory pressure)

num_blocks=16 , block_size=16 → **256** tokens capacity. 10 seqs × 64 tokens
→ **640** needed (4 blocks each). **Predict:** how many run concurrently? Any
preemption?

```
In [37]: exp2_engine = run_exp2_memory_pressure(seed=SEED)
```


=====

Experiment 2: Memory Pressure & Preemption

=====

num_blocks=16, 10 long sequences (64 tokens each) -> watch preemption

```
[metrics] step=1 prefill=19ms decode=0ms gen_tokens=3 tok/s=155 running=3
waiting=7 swapped=0 kv=[██████████] 94% cpu=1%
  step 1 | running= 3 | swapped= 0 | util=94%
[preempt] group 2 -> swapped (blocks freed)
[metrics] step=2 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=3%
  step 2 | running= 2 | swapped= 1 | util=62%
[metrics] step=3 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 3 | running= 2 | swapped= 1 | util=62%
[metrics] step=4 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 4 | running= 2 | swapped= 1 | util=62%
[metrics] step=5 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 5 | running= 2 | swapped= 1 | util=62%
[metrics] step=6 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 6 | running= 2 | swapped= 1 | util=62%
[metrics] step=7 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 7 | running= 2 | swapped= 1 | util=62%
[metrics] step=8 prefill=0ms decode=10ms gen_tokens=2 tok/s=196 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 8 | running= 2 | swapped= 1 | util=62%
[metrics] step=9 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=7 swapped=1 kv=[██████████] 62% cpu=0%
  step 9 | running= 2 | swapped= 1 | util=62%
[metrics] step=10 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=0
waiting=7 swapped=1 kv=[██████████] 0% cpu=0%
  step 10 | running= 0 | swapped= 1 | util=0%
[metrics] step=11 prefill=19ms decode=0ms gen_tokens=3 tok/s=155 running=3
waiting=4 swapped=1 kv=[██████████] 94% cpu=0%
  step 11 | running= 3 | swapped= 1 | util=94%
[preempt] group 5 -> swapped (blocks freed)
[metrics] step=12 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=0%
  step 12 | running= 2 | swapped= 2 | util=62%
[metrics] step=13 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=0%
  step 13 | running= 2 | swapped= 2 | util=62%
[metrics] step=14 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=3%
  step 14 | running= 2 | swapped= 2 | util=62%
[metrics] step=15 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=0%
  step 15 | running= 2 | swapped= 2 | util=62%
[metrics] step=16 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=0%
  step 16 | running= 2 | swapped= 2 | util=62%
[metrics] step=17 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
```

```
waiting=4 swapped=2 kv=[██████████] 62% cpu=0%
  step 17 | running= 2 | swapped= 2 | util=62%
  [metrics] step=18 prefill=0ms decode=11ms gen_tokens=2 tok/s=185 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=3%
  step 18 | running= 2 | swapped= 2 | util=62%
  [metrics] step=19 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=4 swapped=2 kv=[██████████] 62% cpu=2%
  step 19 | running= 2 | swapped= 2 | util=62%
  [metrics] step=20 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=0
waiting=4 swapped=2 kv=[██████████] 0% cpu=0%
  step 20 | running= 0 | swapped= 2 | util=0%
  [metrics] step=21 prefill=19ms decode=0ms gen_tokens=3 tok/s=155 running=3
waiting=1 swapped=2 kv=[██████████] 94% cpu=0%
  step 21 | running= 3 | swapped= 2 | util=94%
  [preempt] group 8 -> swapped (blocks freed)
  [metrics] step=22 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 22 | running= 2 | swapped= 3 | util=62%
  [metrics] step=23 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 23 | running= 2 | swapped= 3 | util=62%
  [metrics] step=24 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 24 | running= 2 | swapped= 3 | util=62%
  [metrics] step=25 prefill=0ms decode=10ms gen_tokens=2 tok/s=194 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 25 | running= 2 | swapped= 3 | util=62%
  [metrics] step=26 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 26 | running= 2 | swapped= 3 | util=62%
  [metrics] step=27 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 27 | running= 2 | swapped= 3 | util=62%
  [metrics] step=28 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 28 | running= 2 | swapped= 3 | util=62%
  [metrics] step=29 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=1 swapped=3 kv=[██████████] 62% cpu=0%
  step 29 | running= 2 | swapped= 3 | util=62%
  [metrics] step=30 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=0
waiting=1 swapped=3 kv=[██████████] 0% cpu=3%
  step 30 | running= 0 | swapped= 3 | util=0%
  [swap-in] group 2 <- swapped
  [metrics] step=31 prefill=7ms decode=5ms gen_tokens=2 tok/s=172 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 31 | running= 2 | swapped= 2 | util=62%
  [metrics] step=32 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=3%
  step 32 | running= 2 | swapped= 2 | util=62%
  [metrics] step=33 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 33 | running= 2 | swapped= 2 | util=62%
  [metrics] step=34 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 34 | running= 2 | swapped= 2 | util=62%
  [metrics] step=35 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
```

```

waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 35 | running= 2 | swapped= 2 | util=62%
  [metrics] step=36 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 36 | running= 2 | swapped= 2 | util=62%
  [metrics] step=37 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 37 | running= 2 | swapped= 2 | util=62%
  [metrics] step=38 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=2 kv=[██████████] 62% cpu=0%
  step 38 | running= 2 | swapped= 2 | util=62%
  [metrics] step=39 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=1
waiting=0 swapped=2 kv=[██████████] 31% cpu=0%
  step 39 | running= 1 | swapped= 2 | util=31%
  [swap-in] group 5 <- swapped
  [metrics] step=40 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=1
waiting=0 swapped=1 kv=[██████████] 31% cpu=0%
  step 40 | running= 1 | swapped= 1 | util=31%
  [swap-in] group 8 <- swapped
  [metrics] step=41 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=3%
  step 41 | running= 2 | swapped= 0 | util=62%
  [metrics] step=42 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 42 | running= 2 | swapped= 0 | util=62%
  [metrics] step=43 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 43 | running= 2 | swapped= 0 | util=62%
  [metrics] step=44 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 44 | running= 2 | swapped= 0 | util=62%
  [metrics] step=45 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 45 | running= 2 | swapped= 0 | util=62%
  [metrics] step=46 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 46 | running= 2 | swapped= 0 | util=62%
  [metrics] step=47 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=2
waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
  step 47 | running= 2 | swapped= 0 | util=62%
  [metrics] step=48 prefill=0ms decode=10ms gen_tokens=2 tok/s=198 running=1
waiting=0 swapped=0 kv=[██████████] 31% cpu=0%
  step 48 | running= 1 | swapped= 0 | util=31%
  [metrics] step=49 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
  step 49 | running= 0 | swapped= 0 | util=0%

```

Done in 49 steps.

```

[metrics] summary:
  prompt_tokens_total=640 generation_tokens_total=100
  time_to_first_token_avg_ms=157 tpot_avg_ms=10 e2e_request_latency_avg_ms
=336
  prompt_len_avg=64 (prefix tokens at start)
  prefill_latency_avg_ms=16

```



```
decode_latency_avg_ms=10
```

Predict — Exp 3 (prefix sharing)

5 seqs, 32-token shared prefix, `block_size=16` → prefix = 2 blocks. **Predict:**
blocks with vs without sharing?

```
In [38]: run_exp3_prefix_sharing()
```

```
=====
Experiment 3: Prefix Sharing (copy_on_write)
=====
5 sequences with same 32-token prefix -> compare utilization

Without sharing – block tables:
  seq 0: [0, 1, 2]
  seq 1: [3, 4, 5]
  seq 2: [6, 7, 8]
  seq 3: [9, 10, 11]
  seq 4: [12, 13, 14]
  ref_counts: {0: 1, 1: 1, 2: 1, 3: 1, 4: 1, 5: 1, 6: 1, 7: 1, 8: 1, 9: 1, 10: 1, 11: 1, 12: 1, 13: 1, 14: 1}

Without prefix sharing: 5 seqs × 3 blocks = 15 blocks -> util=23%
With copy_on_write: shared prefix -> util=11%

With sharing – block tables (note shared block IDs):
  seq 0: [15, 16, 17]
  seq 1: [15, 16, 17, 18]
  seq 2: [15, 16, 17, 19]
  seq 3: [15, 16, 17, 20]
  seq 4: [15, 16, 17, 21]
  ref_counts: {15: 5, 16: 5, 17: 5, 18: 1, 19: 1, 20: 1, 21: 1}
  Shared prefix block IDs: [15, 16]
```

4 — Break it

Each cell: run, then explain which limit bound you.

```
In [39]: # 4.1 Block pool: 8 blocks, 64-token prompts need 4 blocks each → 2 fit, 3rd seq needs 4 -> OOM
from smol_vllm import BlockSpaceManager

bm = BlockSpaceManager(num_blocks=8, block_size=16)
for sid in [0, 1]:
    bm.allocate(sid, 64)
    print(f"seq {sid}: table={bm.get_block_table(sid)}")
print(f"After 2 seqs: free={bm.num_free_blocks()}/8 (3rd seq would need 4 -> OOM)")

# Same budget via scheduler: admission keeps headroom → often running=1 (see scheduler.py)
```

```
print("\nScheduler admission (free_after >= running rule):")
break1 = LLMEngine(num_gpu_blocks=8, block_size=16, max_batch_size=8, seed=5)
for i in range(3):
    break1.add_request(list(range(1000 + i, 1064 + i)), max_tokens=5)
for step in range(8):
    break1.step()
    show_engine_state(break1, step=step)
```

seq 0: table=[0, 1, 2, 3]

seq 1: table=[4, 5, 6, 7]

After 2 seqs: free=0/8 (3rd seq would need 4 – queue or reject)

Scheduler admission (free_after >= running rule):

[metrics] step=1 prefill=7ms decode=0ms gen_tokens=1 tok/s=151 running=1 waiting=2 swapped=0 kv=[██████████] 62% cpu=3%

step 0 | waiting=[1, 2] running=[0] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=2 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 waiting=2 swapped=0 kv=[██████████] 62% cpu=0%

step 1 | waiting=[1, 2] running=[0] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=3 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 waiting=2 swapped=0 kv=[██████████] 62% cpu=0%

step 2 | waiting=[1, 2] running=[0] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=4 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 waiting=2 swapped=0 kv=[██████████] 62% cpu=0%

step 3 | waiting=[1, 2] running=[0] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=5 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=0 waiting=2 swapped=0 kv=[██████████] 0% cpu=0%

step 4 | waiting=[1, 2] running=[] swapped=[] | free=8/8 blocks util=0.0%

[metrics] step=6 prefill=6ms decode=0ms gen_tokens=1 tok/s=154 running=1 waiting=1 swapped=0 kv=[██████████] 62% cpu=0%

step 5 | waiting=[2] running=[1] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=7 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 waiting=1 swapped=0 kv=[██████████] 62% cpu=0%

step 6 | waiting=[2] running=[1] swapped=[] | free=3/8 blocks util=62.5%

[metrics] step=8 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 waiting=1 swapped=0 kv=[██████████] 62% cpu=0%

step 7 | waiting=[2] running=[1] swapped=[] | free=3/8 blocks util=62.5%

In [40]: *# 4.2 block_size trade-off – hold ~128-token capacity, same 64-token prompt*
import math

PROMPT_TOKENS = 64

CAPACITY_TOKENS = 128

for bs **in** [1, 16, 64]:

num_blocks = math.ceil(CAPACITY_TOKENS / bs)

e = LLMEngine(num_gpu_blocks=num_blocks, block_size=bs, max_batch_size=1)

e.add_request(list(range(PROMPT_TOKENS)), max_tokens=4)

```

e.step()
table = e.block_manager.get_block_table(0)
blocks_used = len(table)
slots = blocks_used * bs
wasted = slots - PROMPT_TOKENS
print(
    f"block_size={bs:2d} blocks={num_blocks:3d} table_len={blocks_used:3d} wasted_slots={wasted:2d} table={table}"
)

```

```

[metrics] step=1 prefill=6ms decode=0ms gen_tokens=1 tok/s=154 running=1 waiting=0 swapped=0 kv=[██████████] 51% cpu=2%
block_size= 1 blocks=128 table_len=65 wasted_slots= 1 table=[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64]

```

```

[metrics] step=1 prefill=6ms decode=0ms gen_tokens=1 tok/s=154 running=1 waiting=0 swapped=0 kv=[██████████] 62% cpu=0%
block_size=16 blocks= 8 table_len= 5 wasted_slots=16 table=[0, 1, 2, 3, 4]

```

```

[metrics] step=1 prefill=6ms decode=0ms gen_tokens=1 tok/s=154 running=1 waiting=0 swapped=0 kv=[██████████] 100% cpu=0%
block_size=64 blocks= 2 table_len= 2 wasted_slots=64 table=[0, 1]

```

```

In [41]: # 4.3 max_batch_size=64 but num_gpu_blocks=24 – which limit binds?
break3 = LLMEngine(num_gpu_blocks=24, block_size=16, max_batch_size=64, seed=1234)
for i in range(12):
    break3.add_request(list(range(50)), max_tokens=10)
for step in range(6):
    break3.step()
    print(f"step {step+1}: running={len(break3.scheduler.running)} waiting={len(break3.scheduler.waiting)}")

```

```

[metrics] step=1 prefill=20ms decode=0ms gen_tokens=4 tok/s=199 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=1%
step 1: running=4 waiting=8

```

```

[metrics] step=2 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=0%
step 2: running=4 waiting=8

```

```

[metrics] step=3 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=2%
step 3: running=4 waiting=8

```

```

[metrics] step=4 prefill=0ms decode=23ms gen_tokens=4 tok/s=171 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=1%
step 4: running=4 waiting=8

```

```

[metrics] step=5 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=3%
step 5: running=4 waiting=8

```

```

[metrics] step=6 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4 waiting=8 swapped=0 kv=[██████████] 67% cpu=0%
step 6: running=4 waiting=8

```

5 — Debug it — preemption livelock

1. Run with `preempt_guard=False` — watch thrash
2. Ask: preempting to make room for *whom*?
3. Open `scheduler.py` — find `num_free_blocks() < len(running)`
4. Re-run with `preempt_guard=True` (default)

In [42]: `reproduce_preemption_livelock(seed=SEED)`

```
=====
Debug: preemption livelock (preempt_guard=False)
=====
[metrics] step=1 prefill=36ms decode=0ms gen_tokens=1 tok/s=28 running=1 w
aiting=0 swapped=0 kv=[██████████] 92% cpu=2%
[metrics] step=2 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 92% cpu=6%
[metrics] step=3 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=4 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=5 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=6 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=7 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=8 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=9 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=10 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=11 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[preempt] group 0 -> swapped (blocks freed)
[swap-in] group 0 <- swapped
[metrics] step=12 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
```

Out[42]: `<smol_vllm.engine.LLMEngine at 0x766c71a42410>`

```
In [43]: # Fixed – default preempt_guard=True
fixed = LLMEngine(num_gpu_blocks=24, block_size=16, max_batch_size=4, seed=SEED)
fixed.add_request(list(range(350)), max_tokens=12)
for step in range(20):
    fixed.step()
show_engine_state(fixed)
```

```
[metrics] step=1 prefill=35ms decode=0ms gen_tokens=1 tok/s=28 running=1 w
aiting=0 swapped=0 kv=[██████████] 92% cpu=1%
[metrics] step=2 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 92% cpu=0%
[metrics] step=3 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=4 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=5 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=6 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=7 prefill=0ms decode=11ms gen_tokens=1 tok/s=95 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=6%
[metrics] step=8 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=4%
[metrics] step=9 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1 w
aiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=10 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=11 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
[metrics] step=12 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
waiting=[] running=[] swapped=[] | free=24/24 blocks util=0.0%
```

6 — Throughput & simulator fidelity (Exp 4)

Run twice: `timing="naive"` then `timing="roofline"`. Which cost model can teach continuous batching?

```
In [44]: naive_engines = run_exp4_throughput_scaling(timing="naive", seed=SEED)
roof_engines = run_exp4_throughput_scaling(timing="roofline", seed=SEED)
```

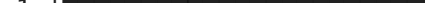
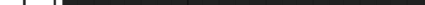
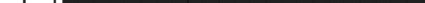
Experiment 4: Throughput Scaling (timing=naive)

[naive] cost $\propto$ batch_size only – sleep($0.005 \times \text{batch_size}$) per decode step (no fixed overhead)

Measure tok/s at batch_size=1, 8, 16 -> ASCII bar chart
(step metrics suppressed here so the bar chart stays visible)

```
batch_size= 1: 196.3 tok/s
batch_size= 8: 198.6 tok/s
[preempt] group 11 -> swapped (blocks freed)
[preempt] group 10 -> swapped (blocks freed)
batch_size=16: 199.2 tok/s
```

Throughput (tok/s) bar chart:

batch= 1		196.3
batch= 8		198.6
batch=16		199.2

```
[metrics] summary:
```

```
prompt_tokens_total=0 generation_tokens_total=0
time_to_first_token_avg_ms=0 tpot_avg_ms=0 e2e_request_latency_avg_ms=0
prompt_len_avg=0 (prefix tokens at start)
```

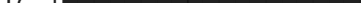
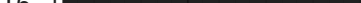
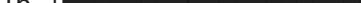
Experiment 4: Throughput Scaling (timing=roofline)

[roofline] fixed + marginal – sleep($0.02 + 0.0005 \times \text{batch_size}$) per decode step

Measure tok/s at batch_size=1, 8, 16 -> ASCII bar chart
(step metrics suppressed here so the bar chart stays visible)

```
batch_size= 1: 50.4 tok/s
batch_size= 8: 320.4 tok/s
[preempt] group 11 -> swapped (blocks freed)
[preempt] group 10 -> swapped (blocks freed)
batch_size=16: 415.7 tok/s
```

Throughput (tok/s) bar chart:

batch= 1		50.4
batch= 8		320.4
batch=16		415.7

```
[metrics] summary:
```

```
prompt_tokens_total=0 generation_tokens_total=0
time_to_first_token_avg_ms=0 tpot_avg_ms=0 e2e_request_latency_avg_ms=0
prompt len avg=0 (prefix tokens at start)
```


Skip if GPU_OK is false. Derive slide numbers from your own [metrics] lines.

```
In [45]: if GPU_OK:
         real_engine = run_exp5_fake_vs_real(seed=SEED)
       else:
         print("GPU track skipped – see LAMBDA.md")
```

```
=====
Experiment 5: Fake vs Real Model (Educational)
=====
```

```
=== Fake model (simulated timing, zero deps) ===
```

```
[metrics] step=1 prefill=8ms decode=0ms gen_tokens=4 tok/s=494 running=4 w
aiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=2 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=3 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=4 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=5 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=6 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=7 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=8 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=0%
[metrics] step=9 prefill=0ms decode=24ms gen_tokens=4 tok/s=168 running=4
waiting=0 swapped=0 kv=[██████████] 6% cpu=3%
[metrics] step=10 prefill=0ms decode=20ms gen_tokens=4 tok/s=199 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=2%
4 prompts, ~20 tok each, max 10 output → 0.20s, 204 tok/s (simulated)
```

```
=== CausalLM (actual compute + memory) ===
```

```
[smol-vllm] Loading TinyLlama/TinyLlama-1.1B-Chat-v1.0 on cuda ...
Loading weights: 0%|          | 0/201 [00:00<?, ?it/s]
```

```

[smol-vllm] Loaded in 1.2s
[edu] Prefill batch=2 compute-bound prompt_tokens≈6 26ms → [13, 2]
[metrics] step=1 prefill=28ms decode=0ms gen_tokens=2 tok/s=73 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=9%
[edu] Decode batch=2 memory-bound KV cache reads 42ms → [13, 29871]
[metrics] step=2 prefill=0ms decode=42ms gen_tokens=2 tok/s=48 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 39ms → [29961, 13]
[metrics] step=3 prefill=0ms decode=39ms gen_tokens=2 tok/s=51 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 40ms → [10858, 29966]
[metrics] step=4 prefill=0ms decode=40ms gen_tokens=2 tok/s=50 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 40ms → [4408, 29989]
[metrics] step=5 prefill=0ms decode=40ms gen_tokens=2 tok/s=51 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 40ms → [29962, 465]
[metrics] step=6 prefill=0ms decode=40ms gen_tokens=2 tok/s=50 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 40ms → [13, 22137]
[metrics] step=7 prefill=0ms decode=40ms gen_tokens=2 tok/s=50 running=2 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=2 memory-bound KV cache reads 39ms → [13, 29989]
[metrics] step=8 prefill=0ms decode=39ms gen_tokens=2 tok/s=51 running=0 waiting=0 swapped=0 kv=[██████████] 0% gpu_mem=6319/6356MB cpu=3%
2 prompts → 0.31s, 51.0 tok/s (real)
VRAM peaked at 6320 MB

[metrics] summary:
prompt_tokens_total=13 generation_tokens_total=16
time_to_first_token_avg_ms=28 tpot_avg_ms=40 e2e_request_latency_avg_ms=313
prompt_len_avg=6 (prefix tokens at start)
prefill_latency_avg_ms=28
decode_latency_avg_ms=40

```

```

In [46]: # Prefill vs decode asymmetry + KV bytes/token (when GPU_OK)
if GPU_OK:
    m = real_engine.metrics
    if m.prefill_latencies and m.decode_latencies:
        p = sum(m.prefill_latencies)/len(m.prefill_latencies)*1000
        d = sum(m.decode_latencies)/len(m.decode_latencies)*1000
        print(f"prefill_avg={p:.0f}ms decode_avg={d:.0f}ms ratio={p/max(d,
    try:
        import torch
        if torch.cuda.is_available():
            mb = torch.cuda.max_memory_allocated()/1024**2
            print(f"VRAM peak={mb:.0f} MB (TinyLlama ~1.1B × 2B ≈ 2.2 GB weights)
        except ImportError:
            pass
    measure_kv_bytes_per_token(seed=SEED)

```

```

prefill_avg=28ms decode_avg=40ms ratio=0.7x
VRAM peak=6320 MB (TinyLlama ~1.1B × 2B ≈ 2.2 GB weights)
[smol-vllm] Loading TinyLlama/TinyLlama-1.1B-Chat-v1.0 on cuda ...

```

Loading weights: 0%| | 0/201 [00:00<?, ?it/s]

```
[smol-vllm] Loaded in 1.4s
[edu] Prefill batch=1 compute-bound prompt_tokens≈18 22ms → [13]
  [metrics] step=1 prefill=23ms decode=0ms gen_tokens=1 tok/s=44 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=8%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [13]
  [metrics] step=2 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29906]
  [metrics] step=3 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29889]
  [metrics] step=4 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [4803]
  [metrics] step=5 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [263]
  [metrics] step=6 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [7090]
  [metrics] step=7 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29899]
  [metrics] step=8 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [711]
  [metrics] step=9 prefill=0ms decode=19ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [17843]
  [metrics] step=10 prefill=0ms decode=20ms gen_tokens=1 tok/s=49 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6319/6356MB cpu=5%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [2738]
  [metrics] step=11 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [2948]
  [metrics] step=12 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29901]
  [metrics] step=13 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=5%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [323]
  [metrics] step=14 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [4901]
  [metrics] step=15 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29931]
  [metrics] step=16 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29880]
  [metrics] step=17 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [3304]
  [metrics] step=18 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1 waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [508]
```

```

[metrics] step=19 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [671]
[metrics] step=20 prefill=0ms decode=20ms gen_tokens=1 tok/s=49 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [263]
[metrics] step=21 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [7090]
[metrics] step=22 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29899]
[metrics] step=23 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [711]
[metrics] step=24 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [17843]
[metrics] step=25 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [2738]
[metrics] step=26 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [2948]
[metrics] step=27 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29892]
[metrics] step=28 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [607]
[metrics] step=29 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [2794]
[metrics] step=30 prefill=0ms decode=20ms gen_tokens=1 tok/s=49 running=1
waiting=0 swapped=0 kv=[██████████] 2% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [393]
[metrics] step=31 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 3% gpu_mem=6320/6356MB cpu=3%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [372]
[metrics] step=32 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=0
waiting=0 swapped=0 kv=[██████████] 0% gpu_mem=6319/6356MB cpu=3%
KV measured ≈ 0.0 KB/token (32 tokens)
Class 3 slide 41 expects ≈ 22 KB/token (2×L×H×D×bytes)

```

8 — Real traffic — CrewAI through your engine

What this section is for: Class 2 showed inference *from the outside* (HTTP, gateways).

Everything until now was *inside the engine* (scheduler, blocks, prefill/decode).

Here we connect both: a **real agent framework** calls **your** engine — same

`add_request` → `step` loop, but the prompts come from CrewAI agents instead of you typing token lists.

Watch the logs, not the final prose. CrewAI wraps the engine; every agent LLM call becomes an `ENGINE REQUEST #N` with the same queues you traced in §1.

Two layers — same stack as production

Layer	Job	In this notebook	What you see in logs
Orchestration (CrewAI)	Break a user task into steps; route work between agents; pass context task → task	Research Analyst → Technical Writer	CrewAI task/agent banners (green/purple UI)
Inference (smol_vllm)	Schedule requests; allocate KV blocks; run prefill/decode; return tokens	One shared LLMEngine for the whole crew	ENGINE REQUEST #1 , #2 , waiting / running / swapped , [metrics]
Model	Turn tokens into next tokens	FakeModel <code>output_mode="text"</code> on CPU (optional TinyLlama on GPU)	Readable fragment vs real answer

§1 manual vs §8 CrewAI — same engine, different caller

	§1 — You drive	§8 — CrewAI drives
Trigger	<code>engine.add_request(...)</code> in a cell	Agent completes a task → <code>SmolVLLMCrewLLM.call()</code>
Calls per user question	1 (or 3 in the walkthrough)	2 — research facts, then write answer
Prompt content	<code>list(range(...))</code> token IDs	Real system/user text from agent roles
Your job in class	Step through diagram by hand	Point at #1 and #2 — <i>same scheduler, new traffic pattern</i>

Uses `output_mode="text"` so FakeModel output looks like words (not `<450>`). For a real answer, run the optional GPU cell at the end of this section.

```
In [47]: from agent_demo import build_crew, kickoff_crew

agent_engine = LLMEngine(
    num_gpu_blocks=24,
```

```

    block_size=16,
    max_batch_size=4,
    enable_metrics=True,
    seed=SEED,
    output_mode="text",
)
crew = build_crew(agent_engine, verbose_engine=True, dual_agent=True)
user_query = "What is paged attention in LLM inference?"
print(f"USER: {user_query}")
print("\nExpect: ENGINE REQUEST #1 (Research Analyst) → #2 (Technical Writer)

```

USER: What is paged attention in LLM inference?

Expect: ENGINE REQUEST #1 (Research Analyst) → #2 (Technical Writer)

```

In [48]: result = await kickoff_crew(crew, {"user_query": user_query})
print(result.raw)
agent_engine.metrics.print_summary()

```

 Crew Execution Started

Crew Execution Started

Name: crew

ID: 05f48851-68cb-4643-9387-5d0dd0bbbd06

 Task Started

Task Started

Name: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

ID: flde0be5-fe4c-4664-b52a-b946b4825c07

=====

ENGINE REQUEST #1 – CrewAI agent: Research Analyst

=====

[engine] prompt truncated to 348 tokens (KV block budget)

prompt (348 tokens): "system: You are Research Analyst. You gather bullet-point facts. Every LLM call goes through smol_vllm.\nYour personal goal is: Extract key facts about the user's topic via the inference engine.\nuser: ..."

[engine] request queued – scheduler state:

waiting=[0] running=[] swapped=[] | free=24/24 blocks util=0.0%

 Agent Started

Agent: **Research Analyst**

Task: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

[metrics] step=1 prefill=35ms decode=0ms gen_tokens=1 tok/s=**29** running=1 waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 1 | waiting=[] running=[0] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=2 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 2 | waiting=[] running=[0] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=3 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 3 | waiting=[] running=[0] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=4 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 4 | waiting=[] running=[0] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=5 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 5 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=6 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 6 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=7 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 7 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=8 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 8 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=9 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 9 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=10 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 10 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=11 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 11 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=12 prefill=0ms decode=5ms gen_tokens=1 tok/s=**196** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 12 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=13 prefill=0ms decode=5ms gen_tokens=1 tok/s=**195** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=6%
step 13 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=14 prefill=0ms decode=6ms gen_tokens=1 tok/s=**168** running=1 waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 14 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.8%

```
[metrics] step=15 prefill=0ms decode=9ms gen_tokens=1 tok/s=109 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=7%
step 15 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=16 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=6%
step 16 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=17 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 17 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=18 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 18 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=19 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 19 | waiting=[] running=[0] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=20 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
step 20 | waiting=[] running=[] swapped=[] | free=24/24 blocks util=0.0%

[engine] finished – output tokens: [99, 104, 105, 110, 103, 32, 105, 110, 3
2, 108, 97, 114, 103, 101, 32, 108, 97, 110, 103, 117]
[engine] decoded text: 'ching in large langu'
```

[Finalize] todos_count=0, todos_with_results=0

✓ Agent Final Answer –

Agent: **Research Analyst**

Final Answer:

ching in large langu

📋 Task Completion —

Task Completed

Name: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

Agent: Research Analyst

Task Started

Name: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-se
ID: 63014b4f-764e-4c1d-8224-73d572bcefe0

=====

ENGINE REQUEST #2 – CrewAI agent: Technical Writer

=====

[engine] prompt truncated to 348 tokens (KV block budget)
prompt (348 tokens): 'system: You are Technical Writer. You polish facts into plain English. Every LLM call goes through smol_vllm.\nYour personal goal is: Turn research notes into a clear answer via the inference engine.\nu...'

[engine] request queued – scheduler state:
waiting=[1] running=[] swapped=[] | free=24/24 blocks util=0.0%

Agent: **Technical Writer**

Task: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-se

[metrics] step=21 prefill=35ms decode=0ms gen_tokens=1 tok/s=29 running=1
waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 1 | waiting=[] running=[1] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=22 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 2 | waiting=[] running=[1] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=23 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 3 | waiting=[] running=[1] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=24 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 92% cpu=0%
step 4 | waiting=[] running=[1] swapped=[] | free=2/24 blocks util=91.7%

[metrics] step=25 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=6%
step 5 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=26 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 6 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=27 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 7 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=28 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 8 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=29 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 9 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=30 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 10 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=31 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=6%
step 11 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=32 prefill=0ms decode=5ms gen_tokens=1 tok/s=197 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 12 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=33 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 13 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

[metrics] step=34 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 14 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.8%

```
[metrics] step=35 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 15 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=36 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 16 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=37 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 17 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=38 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 18 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=39 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=1
waiting=0 swapped=0 kv=[██████████] 96% cpu=0%
step 19 | waiting=[] running=[1] swapped=[] | free=1/24 blocks util=95.
8%
[metrics] step=40 prefill=0ms decode=5ms gen_tokens=1 tok/s=196 running=0
waiting=0 swapped=0 kv=[██████████] 0% cpu=0%
step 20 | waiting=[] running=[] swapped=[] | free=24/24 blocks util=0.0%

[engine] finished – output tokens: [110, 103, 32, 105, 110, 32, 108, 97, 11
4, 103, 101, 32, 108, 97, 110, 103, 117, 97, 103, 101]
[engine] decoded text: 'ng in large language'
```

[Finalize] todos_count=0, todos_with_results=0

✓ Agent Final Answer –

Agent: **Technical Writer**

Final Answer:
ng in large language

📋 Task Completion —

Task Completed

Name: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-se
Agent: Technical Writer

— Crew Completion —

Crew Execution Completed

Name: crew

ID: 05f48851-68cb-4643-9387-5d0dd0bbbd06

Final Output: ng in large language

ng in large language

```
[metrics] summary:
  prompt_tokens_total=696 generation_tokens_total=40
  time_to_first_token_avg_ms=35 tpot_avg_ms=6 e2e_request_latency_avg_ms=1
44  prompt_len_avg=348 (prefix tokens at start)
    prefill_latency_avg_ms=35
    decode_latency_avg_ms=5
```

Tracing Status

Info: Tracing is disabled.

To enable tracing, do any one of these:

- Set tracing=True in your Crew/Flow code
- Set CREWAI_TRACING_ENABLED=true in your project's .env file
- Run: crewai traces enable

```
In [49]: # Comparison recap – print after the crew run
print("=" * 72)
print("CREWAI vs MANUAL ENGINE – what just happened")
print("=" * 72)
print(f"{'':30} {'$1 manual':^18} {'$8 CrewAI':^18}")
print("-" * 72)
rows = [
    ("Who called add_request()", "You", "SmolVLLMCrewLLM adapter"),
    ("Engine instances", "New engine per demo", "One engine, whole crew"),
    ("LLM calls this run", "1 per generate()", str(agent_engine.request_count)),
    ("Scheduler logs", "show_engine_state()", "ENGINE REQUEST #N + state"),
    ("Orchestration visible?", "No", "Yes (CrewAI tasks/agents)"),
    ("Model on CPU", "FakeModel ids/text", "FakeModel text mode"),
]
for label, manual, crew in rows:
    print(f"{label:30} {manual:^18} {crew:^18}")
print("-" * 72)
print("Takeaway: CrewAI is the control plane *above* the engine – not a repl")
print("Production stack: Agent framework → inference server (vLLM) → GPU mat")
print("=" * 72)
```

=====		
CREWAI vs MANUAL ENGINE – what just happened		
=====		
	\$1 manual	\$8 CrewAI

Who called add_request()	You	SmolVLLMCrewLLM adapter
Engine instances	New engine per demo	One engine, whole crew
LLM calls this run	1 per generate()	2
Scheduler logs	show_engine_state()	ENGINE REQUEST #N + state
Orchestration visible?	No	Yes (CrewAI tasks/agents)
Model on CPU	FakeModel ids/text	FakeModel text mode

Takeaway: CrewAI is the control plane <i>above</i> the engine – not a replacement for it.		
Production stack: Agent framework → inference server (vLLM) → GPU matmuls.		
=====		

```
In [50]: # Optional: CrewAI on GPU (when Lambda/CUDA ready)
if GPU_OK:
    gpu_crew_engine = LLMEngine(
        num_gpu_blocks=128,
        block_size=16,
        max_batch_size=2,
        use_real_model=True,
        enable_metrics=True,
        seed=SEED,
    )
    gpu_crew = build_crew(gpu_crew_engine, verbose_engine=True, dual_agent=True)
    gpu_result = await kickoff_crew(gpu_crew, {"user_query": user_query})
    print(gpu_result.raw)
    gpu_crew_engine.metrics.print_summary()
```

```
[smol-vllm] Loading TinyLlama/TinyLlama-1.1B-Chat-v1.0 on cuda ...
Loading weights: 0%|          | 0/201 [00:00<?, ?it/s]
[smol-vllm] Loaded in 1.1s
```

 Crew Execution Started

Crew Execution Started

Name: crew

ID: e879fec0-7c73-42c5-a24d-3200fbbbf0d

 Task Started

Task Started

Name: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

ID: 14d65fc9-749b-49fe-9f82-1194106379a3

Agent: **Research Analyst**

Task: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

```
=====
ENGINE REQUEST #1 – CrewAI agent: Research Analyst
=====
```

```
prompt (133 tokens): "system: You are Research Analyst. You gather bullet-point facts. Every LLM call goes through smol_vllm.\nYour personal goal is: Extract key facts about the user's topic via the inference engine.\nuser: ..."
```

```
[engine] request queued – scheduler state:
waiting=[0]  running=[]  swapped=[]  | free=128/128 blocks  util=0.0%
```

```
[edu] Prefill batch=1 compute-bound prompt_tokens≈133 25ms → [13]
  [metrics] step=1 prefill=27ms decode=0ms gen_tokens=1 tok/s=37 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=8%
step 1 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [13]
  [metrics] step=2 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=5%
step 2 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [29925]
  [metrics] step=3 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 3 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [4063]
  [metrics] step=4 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 4 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 66ms → [8570]
  [metrics] step=5 prefill=0ms decode=66ms gen_tokens=1 tok/s=15 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=4%
step 5 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [338]
  [metrics] step=6 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 6 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [263]
  [metrics] step=7 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 7 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 22ms → [1134]
  [metrics] step=8 prefill=0ms decode=22ms gen_tokens=1 tok/s=46 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=4%
step 8 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [310]
  [metrics] step=9 prefill=0ms decode=22ms gen_tokens=1 tok/s=46 running=1 w
aiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 9 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [8570]
  [metrics] step=10 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1
waiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 10 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [393]
  [metrics] step=11 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1
waiting=0 swapped=0 kv=[██████████] 7% gpu_mem=6322/6362MB cpu=3%
step 11 | waiting=[] running=[0] swapped=[] | free=119/128 blocks util=
7.0%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [338]
```

```

[metrics] step=12 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=3%
step 12 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [21309]
[metrics] step=13 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=3%
step 13 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [373]
[metrics] step=14 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=5%
step 14 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 25ms → [263]
[metrics] step=15 prefill=0ms decode=25ms gen_tokens=1 tok/s=40 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=3%
step 15 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 24ms → [2702]
[metrics] step=16 prefill=0ms decode=25ms gen_tokens=1 tok/s=41 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=4%
step 16 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 24ms → [11261]
[metrics] step=17 prefill=0ms decode=25ms gen_tokens=1 tok/s=41 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=3%
step 17 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 26ms → [470]
[metrics] step=18 prefill=0ms decode=26ms gen_tokens=1 tok/s=39 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=4%
step 18 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 24ms → [9565]
[metrics] step=19 prefill=0ms decode=24ms gen_tokens=1 tok/s=42 running=1
waiting=0 swapped=0 kv=[██████████] 8% gpu_mem=6322/6362MB cpu=3%
step 19 | waiting=[] running=[0] swapped=[] | free=118/128 blocks util=
7.8%
[edu] Decode batch=1 memory-bound KV cache reads 23ms → [310]
[metrics] step=20 prefill=0ms decode=23ms gen_tokens=1 tok/s=44 running=0
waiting=0 swapped=0 kv=[██████████] 0% gpu_mem=6319/6362MB cpu=4%
step 20 | waiting=[] running=[] swapped=[] | free=128/128 blocks util=0.
0%

```

```

[engine] finished – output tokens: [13, 13, 29925, 4063, 8570, 338, 263, 113
4, 310, 8570, 393, 338, 21309, 373, 263, 2702, 11261, 470, 9565, 310]
[engine] decoded text: '\n\nPaged attention is a type of attention that is f
ocused on a specific topic or aspect of'

```

```

[Finalize] todos_count=0, todos_with_results=0

```

✓ Agent Final Answer —

Agent: **Research Analyst**

Final Answer:

Paged attention is a type of attention that is focused on a specifi

📋 Task Completion —

Task Completed

Name: The user asked: What is paged attention in LLM inference?

Use the inference engine to list 3 key facts about this topic.

Agent: Research Analyst

📋 Task Started —

Task Started

Name: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-se

ID: fd859bf1-cc31-48b7-a1bd-0eb6e0a222ee

=====

ENGINE REQUEST #2 – CrewAI agent: Technical Writer

=====

prompt (173 tokens): 'system: You are Technical Writer. You polish facts into plain English. Every LLM call goes through smol_vllm.\nYour personal goal is: Turn research notes into a clear answer via the inference engine.\nu...'

[engine] request queued – scheduler state:

waiting=[1] running=[] swapped=[] | free=128/128 blocks util=0.0%

🤖 Agent Started —

Agent: **Technical Writer**

Task: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-se

```
[edu] Prefill batch=1 compute-bound prompt_tokens≈173 24ms → [13]
[metrics] step=21 prefill=27ms decode=0ms gen_tokens=1 tok/s=37 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=7%
step 1 | waiting=[] running=[1] swapped=[] | free=117/128 blocks util=
8.6%
[edu] Decode batch=1 memory-bound KV cache reads 21ms → [13]
[metrics] step=22 prefill=0ms decode=21ms gen_tokens=1 tok/s=48 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 2 | waiting=[] running=[1] swapped=[] | free=117/128 blocks util=
8.6%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [29925]
[metrics] step=23 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 3 | waiting=[] running=[1] swapped=[] | free=117/128 blocks util=
8.6%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [4063]
[metrics] step=24 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 4 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [8570]
[metrics] step=25 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 5 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [338]
[metrics] step=26 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 6 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [263]
[metrics] step=27 prefill=0ms decode=19ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 7 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [1134]
[metrics] step=28 prefill=0ms decode=19ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 8 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [310]
[metrics] step=29 prefill=0ms decode=19ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 9 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [8570]
[metrics] step=30 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 10 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [393]
[metrics] step=31 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=5%
step 11 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [338]
```

```

[metrics] step=32 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 12 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 19ms → [21309]
[metrics] step=33 prefill=0ms decode=19ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 13 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [373]
[metrics] step=34 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 14 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [263]
[metrics] step=35 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 15 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [2702]
[metrics] step=36 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 16 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [11261]
[metrics] step=37 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 17 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [470]
[metrics] step=38 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 18 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [9565]
[metrics] step=39 prefill=0ms decode=20ms gen_tokens=1 tok/s=51 running=1
waiting=0 swapped=0 kv=[██████████] 9% gpu_mem=6323/6364MB cpu=3%
step 19 | waiting=[] running=[1] swapped=[] | free=116/128 blocks util=
9.4%
[edu] Decode batch=1 memory-bound KV cache reads 20ms → [310]
[metrics] step=40 prefill=0ms decode=20ms gen_tokens=1 tok/s=50 running=0
waiting=0 swapped=0 kv=[██████████] 0% gpu_mem=6319/6364MB cpu=3%
step 20 | waiting=[] running=[] swapped=[] | free=128/128 blocks util=0.
0%

```

```

[engine] finished – output tokens: [13, 13, 29925, 4063, 8570, 338, 263, 113
4, 310, 8570, 393, 338, 21309, 373, 263, 2702, 11261, 470, 9565, 310]
[engine] decoded text: '\n\nPaged attention is a type of attention that is f
ocused on a specific topic or aspect of'

```

```

[Finalize] todos_count=0, todos_with_results=0

```

✓ Agent Final Answer —

Agent: **Technical Writer**

Final Answer:

Paged attention is a type of attention that is focused on a specific

📋 Task Completion —

Task Completed

Name: The user asked: What is paged attention in LLM inference?

Using the research notes from the previous task, write a clear 2-sentence

Agent: Technical Writer

Paged attention is a type of attention that is focused on a specific topic or aspect of

```
[metrics] summary:
  prompt_tokens_total=306 generation_tokens_total=40
  time_to_first_token_avg_ms=27 tpot_avg_ms=23 e2e_request_latency_avg_ms=
460
  prompt_len_avg=153 (prefix tokens at start)
  prefill_latency_avg_ms=27
  decode_latency_avg_ms=22
```

Crew Completion —

Crew Execution Completed

Name: crew

ID: e879fec0-7c73-42c5-a24d-3200fbbbaf0d

Final Output:

Paged attention is a type of attention that is focused on a specific

Tracing Status —

Info: Tracing is disabled.

To enable tracing, do any one of these:

- Set tracing=True in your Crew/Flow code
- Set CREWAI_TRACING_ENABLED=true in your project's .env file
- Run: crewai traces enable

9 — Diff to production vLLM

Open side by side:

smol_vllm	production
<code>scheduler.py</code> → <code>schedule()</code>	<code>vllm/v1/core/sched/scheduler.py</code>
waiting / running / swapped	same skeleton
allocate + preempt	+ chunked prefill, prefix cache, speculative decode, grammar bitmasks

The toy wasn't a toy — it's the control plane with the extras removed.

Export PDF (optional)

Run after all cells execute.

```
In [51]: from lib.export_notebook import save_pdf
from IPython.display import FileLink

pdf = save_pdf("class5.ipynb")
FileLink(pdf.name)
```

Out[51]: `class5.ipynb.pdf`